

Cyclistic Case Study

Converting Casual Riders into Annual Members

Project Overview

Cyclistic is a bike-sharing company that offers two types of services:

- Casual rides (single ride or day pass)
- Annual membership subscriptions

The business objective of this analysis is to identify key behavioral differences between casual riders and annual members and provide data-driven recommendations to help convert casual riders into annual members.

This project demonstrates:

- Data cleaning and validation
- SQL-based exploratory data analysis
- Behavioral segmentation
- Business-driven insights
- Interactive dashboard development (Tableau)

Tools & Technologies

- **SQL (BigQuery)** – data cleaning & analysis
- **Tableau Public** – dashboard & visualization
- **Excel** – initial structure validation
- **GitHub / Markdown** – documentation

Data Source

The dataset consists of 12 months of historical trip-level data provided by Lyft Bikes and Scooters, LLC under the Divvy Data License Agreement.

Dataset size: 5.5 million ride records

Each record includes:

- Ride ID
- Rideable type
- Start and end timestamps
- Station information
- Coordinates
- Membership type (casual or member)

Data Credibility (ROCCC Framework)

- **Reliable:** Official operational data
- **Original:** Primary system trip data
- **Comprehensive:** Behavioral metrics available
- **Current:** Most recent 12-month period
- **Cited:** Publicly licensed dataset

Limitations

The dataset does not include demographic information, income levels, or motivations. All conclusions are based strictly on observable ride behavior.

Data Cleaning & Preparation

Data cleaning was performed in SQL (BigQuery).

Removing invalid ride durations

```
-- Remove negative ride durations
DELETE FROM cleaned_trips
WHERE ride_length < 0;

-- Remove rides longer than 24 hours (system anomalies)
DELETE FROM cleaned_trips
WHERE ride_length > 1440;
```

Less than 0.2% of records were removed as anomalies.

Zero-minute rides were retained after validation, as they may represent immediate cancellations.

Final dataset: **5,546,356 valid rides**

Feature Engineering

Additional analytical fields were created:

```
-- Extract hour of day
EXTRACT(HOUR FROM started_at) AS hour_of_day

-- Extract day of week
FORMAT_TIMESTAMP('%A', started_at) AS day_of_week

-- Extract month
EXTRACT(MONTH FROM started_at) AS month

-- Calculate ride length in minutes
TIMESTAMP_DIFF(ended_at, started_at, MINUTE) AS ride_length
```

Exploratory Data Analysis

Ride Duration

Metric	Casual	Member
Average Duration	18.65 min	11.51 min
Median Duration	11 min	8 min

Casual riders take significantly longer rides, suggesting recreational usage. Members exhibit shorter, consistent trips aligned with commuting behavior.

Weekly Usage Patterns

- Casual riders peak on Saturdays (20.69%)
- Members show consistent weekday usage

This indicates:

- Casual riders → leisure-oriented
- Members → routine commuting behavior

Hourly Usage Patterns

Members demonstrate clear commuting peaks:

- Morning peak (~8 AM)

- Evening peak (~5 PM)

Casual riders show gradual daytime increases with no strong morning commute pattern.

This strongly supports the transportation vs. leisure segmentation.

Seasonal Trends

- Casual usage peaks in summer (August: 16.9%)
- Member usage remains stable year-round

Casual behavior appears weather-dependent, while members integrate the service into daily routines.

Key Insights

The analysis reveals two fundamentally different behavioral segments:

Annual Members

- Commute-oriented
- Weekday heavy
- Stable year-round usage
- Clear morning & evening peaks

Casual Riders

- Leisure-oriented
- Weekend-heavy usage
- Summer concentration
- Longer average rides

Cyclistic serves two distinct use cases within the same platform.

Recommendations

1) Priority Strategy: Targeted Conversion Discount

Introduce a limited-time discounted annual membership offer targeting high-frequency casual riders during peak summer months and weekends.

These users demonstrate the highest engagement and represent the most conversion-ready segment.

2) Seasonal Membership Plans

Offer summer-focused membership options for leisure-oriented users who show strong seasonal engagement but limited off-season activity.

3) Weekend-Focused Membership

Develop a weekend-only subscription model to create a lower-commitment pathway toward full annual membership.

Interactive Dashboard

View the interactive Tableau dashboard here:

https://public.tableau.com/views/CyclisticCase_17710796605430/BehavioralDifferences?:language=en-US&:sid=&:redirect=auth&:display_count=n&:origin=viz_share_link

The dashboard highlights:

- Hourly commuting peaks
- Weekly usage differences
- Seasonal patterns
- Ride duration comparison

Conclusion

Cyclistic currently serves two fundamentally different behavioral segments.

By targeting highly engaged casual riders during peak usage periods and introducing tiered membership pathways, the company can strategically convert leisure-oriented riders into long-term annual members while maintaining operational stability.

What This Project Demonstrates

- Data cleaning & validation
- Feature engineering
- SQL-based exploratory analysis
- Behavioral segmentation

- Business-focused recommendations
- Dashboard storytelling